Type MS · Angle between two planes using a right-angled triangle

Answer 1 of 11 · Question 9 part whole · 6 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

9	**Be careful to look on the sketch to award marks for the question and for any equivalent calculations or alternative methods.** **IF IN DOUBT SEND TO REVIEW** X is the point directly below V positioned on the base $ABCDE$ Perpendicular from the mid-point of BC (M) to point X	
	$(\angle BXC =)72^\circ$ or $(\angle ABC =)108^\circ$	B1
	Length of BX eg $(BX =)\dfrac{x}{\sin 36^\circ}(=1.701x)$ or $\dfrac{\sin 54}{\sin 72}\times 2x$ or $\dfrac{x}{\cos 54}$ oe	M1
	$(VX =)\sqrt{(3x)^2-("1.701x")^2}=2.471x$ or $(VM =)\sqrt{(3x)^2-(x)^2}=\sqrt{8}x(=2.828x)$	M1
	$(MX =)\sqrt{("1.701x")^2-x^2}=1.376x$ or $\dfrac{x}{\tan 36}$ oe or $x\tan 54$	M1
	Required angle is $\angle VMX$ or $\angle VME$ $(\tan\angle VMX =)\dfrac{2.471x}{1.376x}\Rightarrow(\angle VMX = 60.888...^\circ)$ oe $(\cos\angle VME =)\dfrac{("\sqrt{8}x")^2+("1.701"x+"1.376"x)^2-(3x)^2}{2\times"\sqrt{8}x"\times("1.701"x+"1.376"x)}$ oe $\Rightarrow$ awrt 60.9° or better	dddM1 A1 [6]
ALT1	**May only be applied if they attempt to find MX in one step and attempt to find VM and use triangle VMX. If they find MX any other way, apply one of the other schemes** $(\angle BXC =)72^\circ$ or $(\angle ABC =)108^\circ$	B1
	$(MX =)\ \dfrac{x}{\tan 36}$ oe or $x\tan 54$	M2
	$(VM =)\sqrt{(3x)^2-(x)^2}=\sqrt{8}x(=2.828x)$	M1
	$(\cos\angle VMX =)\dfrac{"x\tan 54"}{"2\sqrt{2}x"}\Rightarrow(\angle VMX = 60.888...^\circ)$	dddM1
	$\Rightarrow$ awrt 60.9° or better	A1 [6]

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ALT2	$(\angle BXC =)72^\circ$ or $(\angle ABC =)108^\circ$	B1
	$(EB =)\sqrt{(2x)^2+(2x)^2-2(2x)(2x)\cos 108}\,(=3.236x)$ or $(EB =)\dfrac{2x}{\sin 36}\times\sin 108(=3.236x)$	M1
	$(VM =)\sqrt{(3x)^2-(x)^2}=\sqrt{8}x(=2.828x)$	M1
	$(EM =)\sqrt{("3.236x")^2-x^2}\,(=3.077x)$	M1
	$(\cos\angle VME =)\dfrac{("\sqrt{8}x")^2+("1.701"x+"1.376"x)^2-(3x)^2}{2\times"\sqrt{8}x"\times("1.701"x+"1.376"x)}$	dddM1
	$\Rightarrow$ awrt 60.9° or better	A1 [6]
		Total 6 marks

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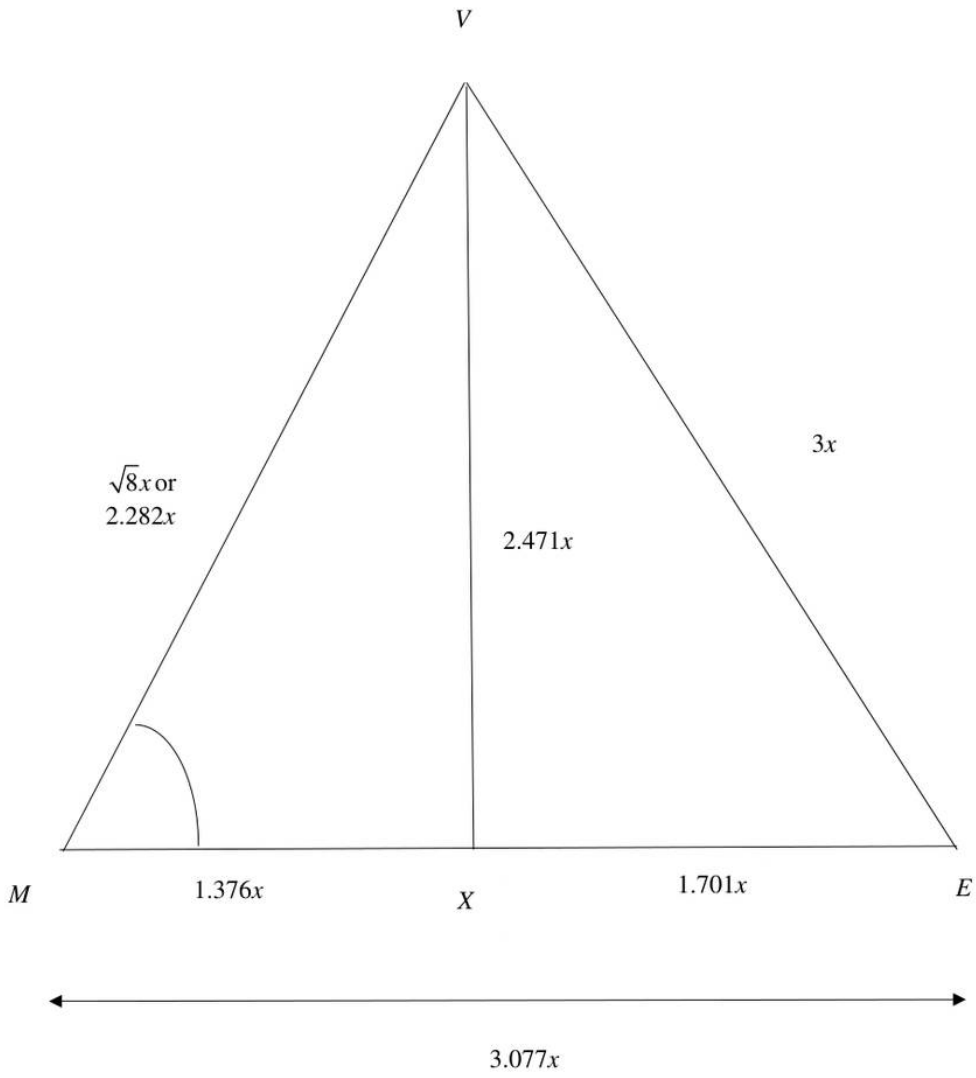
2023 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Mark	Notes
B1	For writing down or finding the angle $BXC = 72°$ or $\angle ABC = 108°$, any notation. Clear identification or implicit use of this angle in later working is acceptable. Note use of 36 degrees or 54 degrees implied this mark.
M1	Ignore missing x's throughout their solution. For **any** correct, suitable trigonometry to find the length BX
M1	Correct use of Pythagoras theorem (with a – sign) or suitable trigonometry to find the height VX of the pyramid or the length VM
M1	Correct use of Pythagoras theorem (with a – sign) or suitable trigonometry to find the length of the midpoint of BC to point X
dddM1	For any suitable trigonometry to find the size of the required angle. Dependent on all 3 previous method marks.
A1	For the correct angle, awrt 60.9°
ALT1 B1	For writing down or finding the angle $BXC = 72°$ or $\angle ABC = 108°$, any notation. Clear identification or implicit use of this angle in later working is acceptable. Note use of 36 degrees or 54 degrees implied this mark.
M2	Ignore missing x's throughout their solution. For **any** correct method to find the length MX
M1	Correct method to find the length VM
dddM1	For any suitable trigonometry to find the size of the required angle. Dependent on all 3 previous method marks.
A1	For the correct angle, awrt 60.9°
ALT2 B1	For writing down or finding the angle $BXC = 72°$ or $\angle ABC = 108°$, any notation. Clear identification or implicit use of this angle in later working is acceptable. Note use of 36 degrees or 54 degrees implied this mark.
M1	Ignore missing x's throughout their solution. For **any** correct method to find the length EB
M1	For any correct method to find VM
M1	For any correct method to find EM
dddM1	For any suitable trigonometry to find the size of the required angle. Dependent on all 3 previous method marks.
A1	For the correct angle, awrt 60.9°
Where students are allowed to use their values, indicated by inverted commas, it MUST be clear from working or their sketch this is the correct use of the relevant side. Beware $\sqrt{8}x$ coming from incorrect working of $\sqrt{(2x)^2 + (2x)^2}$	

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Type MS · Angle between two planes using a right-angled triangle

Answer 1 of 11 · Question 9 part whole · 6 marks

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Question	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

2020 NOV · Paper 01 · extract 1 of 20 · find part c in this mark scheme extract

3(a)	$AM = \sqrt{6^2+8^2} = 10$	M1
	$AE = \sqrt{14^2+10^2} = \sqrt{296} = 17.20... = 17.2\,\text{cm}$	M1A1 (3)
(b)	$\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, $\phi = 54.46... = 54.5°$ or using another trig function	M1A1ft,A1(3)
(c)	$\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8}$, $\theta = 60.255...° = 60.3°$	M1A1ft,A1 (3) [9]

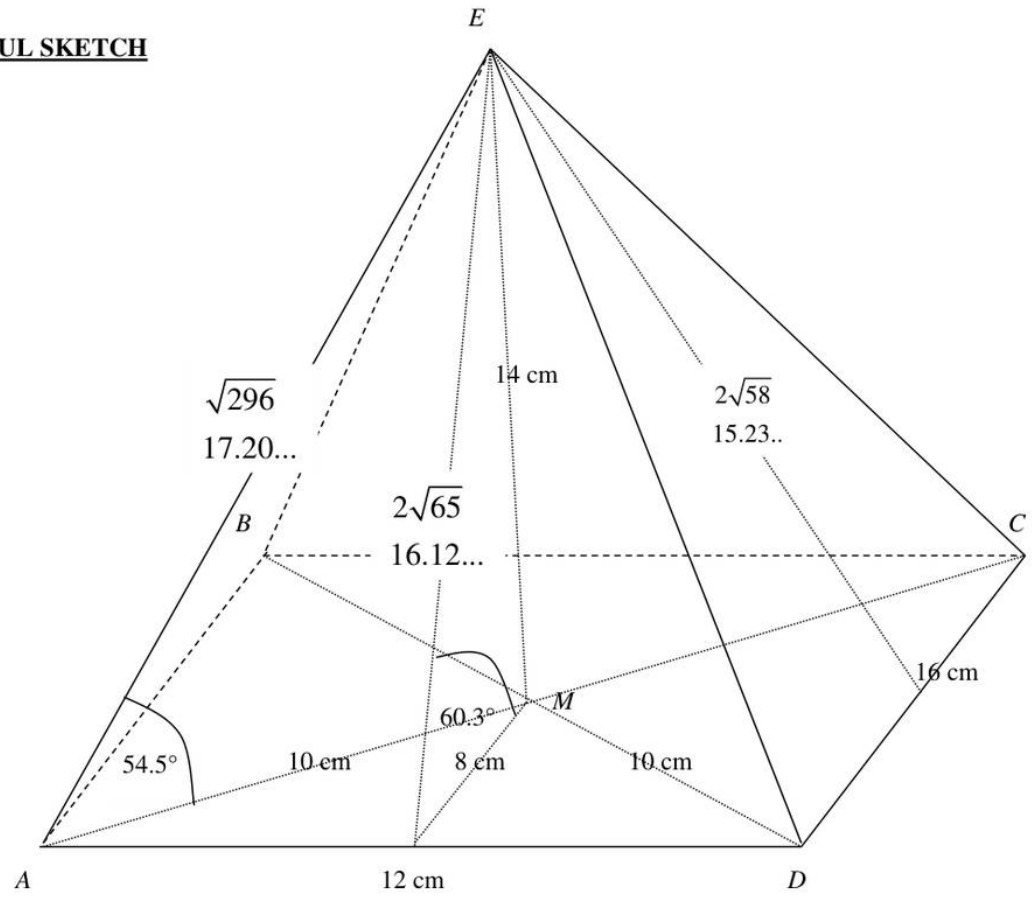
Part	Mark	Notes
(a)	M1	Applies Pythagoras theorem to find the length of AM $AM = \sqrt{6^2+8^2} = 10$ or $AM = \frac{\sqrt{12^2+16^2}}{2} = 10$
	M1	Applies Pythagoras to find the length of one of the sloping edges $AE = \sqrt{14^2+10^2} = \sqrt{296} = ...$
	A1	For the correct length of either AE, DE, CE or BE $AE = 17.2\,\text{cm}$ rounded correctly
	ALT	
	M1M1	Applies Pythagoras in 3D $AE = \sqrt{14^2+6^2+8^2} = \sqrt{296} = ...$
(b)	M1	For applying any acceptable trigonometry to find the required angle. $\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, or $\sin\phi = \frac{14}{\sqrt{296}}$, or $\cos\phi = \frac{10}{\sqrt{296}} \Rightarrow \phi = ...$
	A1ft	For the correct trigonometry if they use sine or cosine following through their $\sqrt{296}$
	A1	Required angle = 54.5° Rounded correctly
(c)	M1	For applying trigonometry to find the required angle. $\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8} \Rightarrow \theta = ...$ OR The length of the perpendicular from E to the mid-point of AD is $\sqrt{260}$ $\sin\theta = \left(\frac{14}{'\sqrt{260}'}\right)$, or $\cos\left(\frac{8}{'\sqrt{260}'}\right) \Rightarrow \theta = ...$
	A1ft	Ft their $\sqrt{260}$
	A1	$\theta = 60.3°$
Rounding: Penalise rounding only the first time it occurs in either (b) or (c)		

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Answer 2 of 11 · Question 3 part c · 3 marks

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USEFUL SKETCH



Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

2020 NOV · Paper 01 · extract 4 of 20 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	Divides through $x^3-4x^2+5=0$ by x^2 and rearranges to achieve as a minimum $x+\frac{5}{x^2}=k$ where k is a constant $\left[x-4+\frac{5}{x^2}=0\Rightarrow x+\frac{5}{x^2}=4\right]$
	M1	**Draw the line $y=k$** following through their value for k No line is M0
	A1	For the two values of $x=1.4$ and $x=3.6$ Condone answers given as coordinates provided they are completely correct. (1.4, 4) and (3.6, 4) Require both M marks for this mark.
(b)	M1	For setting $x+\frac{5}{x^2}=Ax+B\Rightarrow x^3+5=Ax^3+Bx^2\Rightarrow Ax^3-x^3+Bx^2-5=0$, and equating coefficients with x^3-x^2-5 $x^3(A-1)+Bx^2-5\equiv x^3-x^2-5$ to achieve as a minimum $A=(\pm 2)$, $B=(\pm 1)$
	A1	For the correct straight line $y=2x-1$
	ALT – to find the line $y=2x-1$	
	M1	Divides through $x^3-x^2-5=0$ by x^2 and rearranges the equation to achieve as a minimum $\Rightarrow \pm 2x\pm 1=x+\frac{5}{x^2}$
	A1	For the correct straight line $y=2x-1$
	dM1	**Draws their $y=2x-1$ on the graph** and locates the point of intersection. Please check that they draw their line correctly. Coordinates for you to check are (0.5, 0) and (2.5, 4) No line is M0 This mark is dependent on the first M mark in (b)
	A1	For the correct value of $x=2.1$ [allow $x=2.2$] Can only score this mark from M1A1M1 Do not accept the answer given as coordinates.

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
5(a)	Gradient $PR = \frac{6}{-6} = -1$, Gradient $QS = \frac{8}{8} = 1$	M1A1
	Product $= -1 \Rightarrow$ perpendicular	A1 (3)
(b)	(i) $PR = \sqrt{6^2+6^2} = 6\sqrt{2}$ $\left(=\sqrt{72}\right)$	M1
	(ii) $QS = \sqrt{8^2+8^2} = 8\sqrt{2}$ $\left(=\sqrt{128}\right)$	A1 (2)
(c)	Area $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= 48$ $\left(\text{units}^2\right)$	M1A1 (2)
		[7]

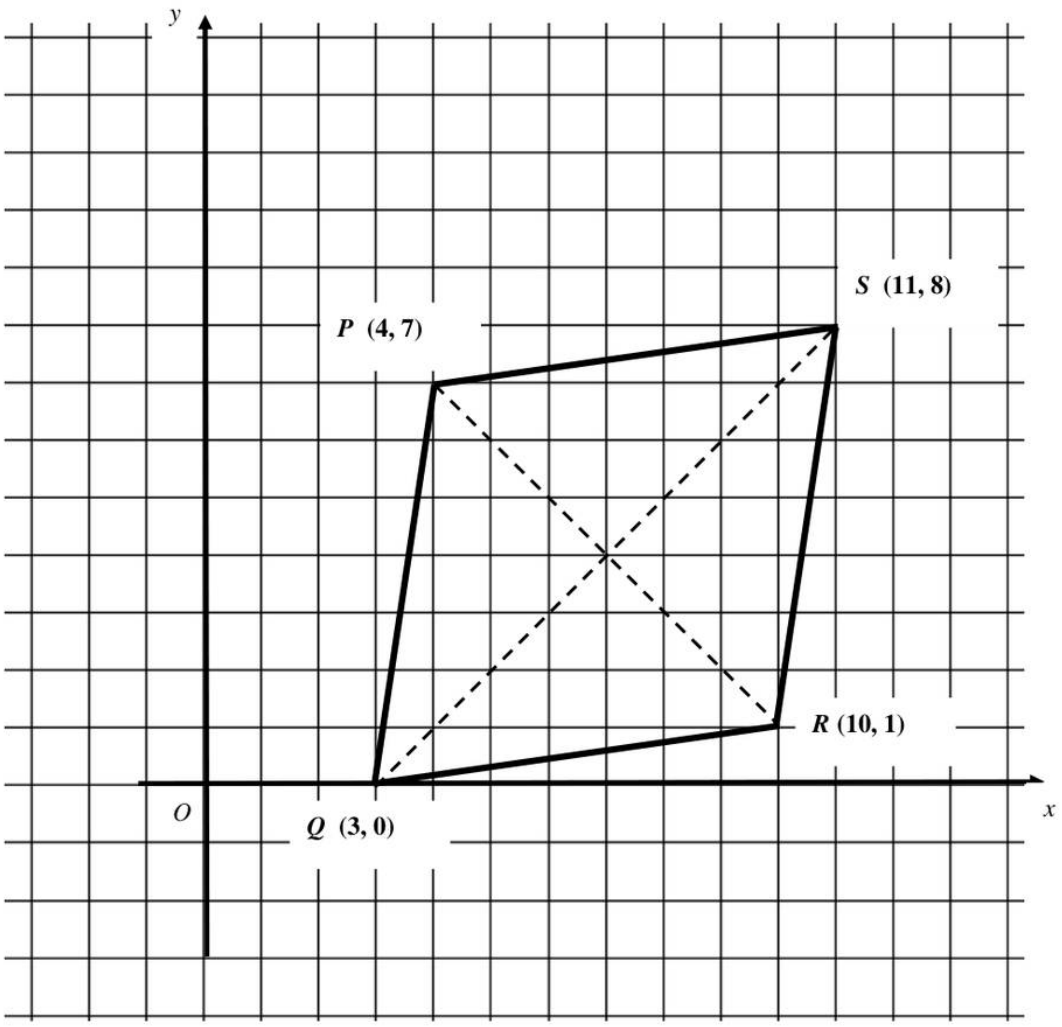
Part	Mark	Notes
(a)	M1	Finds the gradient of PR and QS using a correct method. This may be on a diagram. Gradient $PR = \frac{7-1}{4-10} = \frac{6}{-6} = -1$, Gradient $QS = \frac{8-0}{11-3} = \frac{8}{8} = 1$
	A1	Both gradients correct Gradient $PR = -1$, Gradient $QS = 1$
	A1	Finds the product of the two gradients with a statements that as the product = –1 then the lines are perpendicular.
(b)	M1	For **either** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **OR** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
	A1	For **both** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **AND** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
(c)	M1	$PQRS$ is a kite so $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= \ldots$
	A1	Area = 48 $\left(\text{units}^2\right)$
	ALT Uses determinants	
	M1	Area $= \frac{1}{2}\begin{pmatrix} 4 & 3 & 10 & 11 & 4 \\ 7 & 0 & 1 & 8 & 7 \end{pmatrix}$ $= \frac{1}{2}\left(\left[4\times0+3\times1+10\times8+11\times7\right]-\left[3\times7+10\times0+11\times1+4\times8\right]\right)$ $= \ldots$ Allow one slip in a product.
	A1	Area = 48 $\left(\text{units}^2\right)$

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Answer 2 of 11 · Question 3 part c · 3 marks

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USEFUL SKETCH



Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1
	$2\times17+d=38 \Rightarrow d=4$	A1 (4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1 (2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$	
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1
	$2p^2-11p+12=0$	A1
	$(2p-3)(p-4)=0 \Rightarrow p=4$ $\left(p\neq\frac{3}{2};\ \text{may not be seen}\right)$	M1A1 (4)
		[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $[a = S_1 = 1(15+2\times1) = 17]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}(2a+[n-1]d) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = ...$ and $30 = 2a-4 \Rightarrow a = ...$
		For the correct value of $d = 4$

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Answer 2 of 11 · Question 3 part c · 3 marks

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	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $\mathrm{f}(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $\mathrm{f}(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

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Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

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	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1 (2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$	
	$AC = 6\sqrt{6}$	A1 (4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1 (4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ (or ϕ or $360 + \phi$ (not possible))	M1
	$\therefore \theta = 180 - 2\phi$	A1 (2) [12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for $\cos BAD$ and $\cos BAC$: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

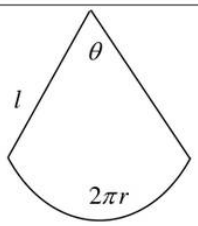
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	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2+x^2-6^2}{2\times6\times x}$ and $\cos BCA = \frac{6^2+(2x)^2-12^2}{2\times6\times2x}$
	M1	$\frac{6^2+x^2-6^2}{2\times6\times x} = \frac{6^2+(2x)^2-12^2}{2\times6\times2x} \Rightarrow x^2 = 2x^2-54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ+\phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ+\phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ+\phi^\circ) = \frac{x\sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ+\phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta+\phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta+\phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ)+\sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta+\phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta+\phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ+\phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ+\phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ+\phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180-\phi)^\circ \Rightarrow \theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$
	A1	For rearranging $\theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$ to achieve $\therefore \theta = 180-2\phi$ This is a show question and there must be no errors here.

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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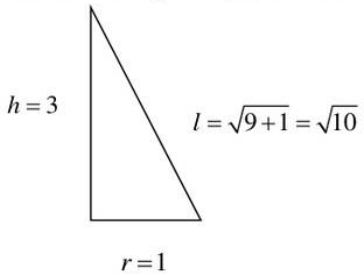
Question Number	Scheme	Marks
10(a)	Circumference of base $= 2\pi r$ 	B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$	M1A1 (4)
(b)	$A = \pi rl$	
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$	B1
	$\dfrac{dA}{dr} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$	M1A1 (3)
(c)	$\dfrac{dV}{dt} = 1.5\ \left(\text{cm}^3/\text{s}\right)$	B1
	$\dfrac{dA}{dt} = \dfrac{dA}{dr} \times \dfrac{dr}{dV} \times \dfrac{dV}{dt}$	M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$	B1
	$\dfrac{dV}{dr} = 3\pi r^2$	B1ft
	$\therefore \dfrac{dA}{dt} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3 \times 64\pi} = 0.3952... = 0.395\ \text{cm}^2/\text{s}$	A1 (5)
		[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2} \times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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Question Number	Scheme	Marks
11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1 (5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1 (4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1
	$\mu = \frac{7}{16}$	A1 (4)
		[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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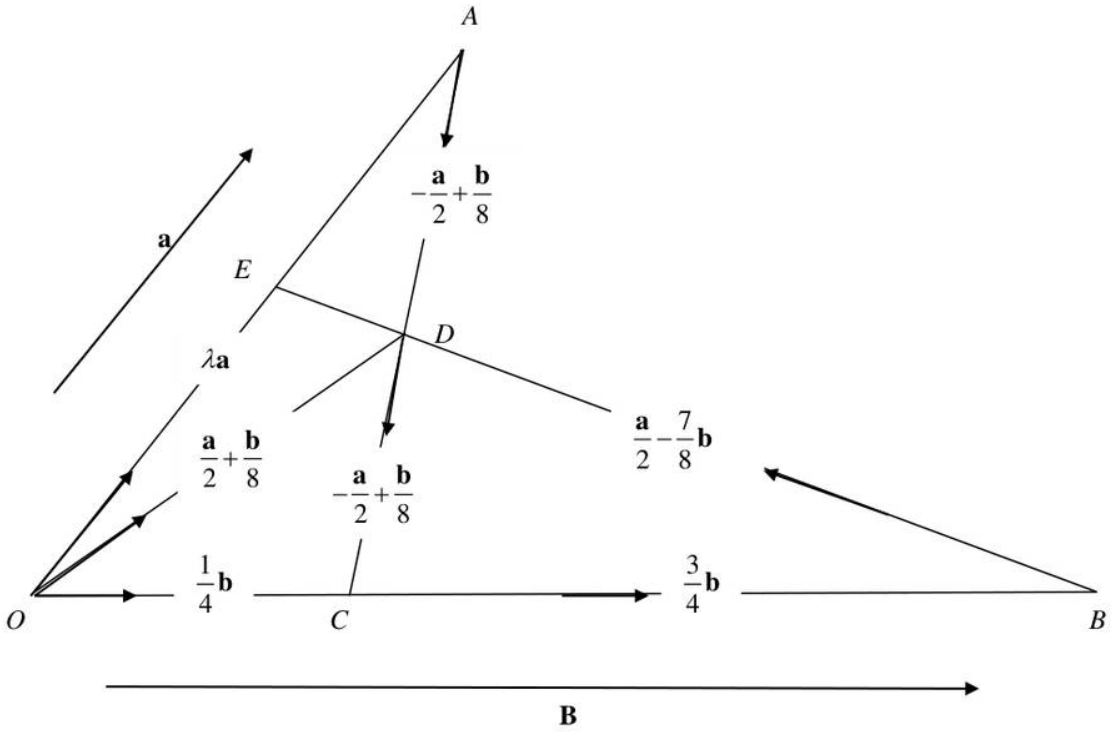
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda$ $\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

Type MS · Angle between two planes using a right-angled triangle

Answer 2 of 11 · Question 3 part c · 3 marks

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USEFUL SKETCH



Type MS · Angle between two planes using a right-angled triangle

Answer 3 of 11 · Question 8 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8 (a)	$BN = 10\cos 60^\circ$ $= 5$ (cm)*	M1 A1cso [2]
(b)	$FN^2 = 10^2 - 5^2 = 75$ or $FN = 10\sin 60 = 5\sqrt{3}$ or $FN = 5\tan 60$ $NE = \sqrt{12^2 + 75} = 14.7986... \approx 14.8$ (cm)	M1 M1A1 [3]
(c)	Let R be the midpoint of FN and P be the point on AD that corresponds to point N on BC. $\angle RPN = \tan^{-1}\left(\dfrac{\sqrt{75}/2}{12}\right) = 19.84167...^\circ \approx 19.8^\circ$	M1A1 [2]
(d)	Let the perpendicular from Y to line BC meet BC at Z $AY = \sqrt{AZ^2 + ZY^2} = \sqrt{12^2 + 6.5^2 + \left(\dfrac{\sqrt{75}}{2}\right)^2} = \sqrt{205}$ $FC = \sqrt{FN^2 + NC^2} = \sqrt{75 + 3^2} = \sqrt{84} \Rightarrow FY = \sqrt{21}$ So $EY = \sqrt{12^2 + 21} = \sqrt{165}$ $\angle AYE = \cos^{-1}\left(\dfrac{205 + 165 - 100}{2 \times \sqrt{205} \times \sqrt{165}}\right) = 42.7745...^\circ \approx 42.8^\circ$	M1A1 M1 M1 M1A1 [6]
		Total 13 marks

(a)	M1	Use cosine or sine in triangle BNF, or any complete method. $BN = 10\cos 60^\circ$ or $\cos 60^\circ = \dfrac{BN}{10}$ or $\dfrac{BN}{\sin 30} = \dfrac{10}{\sin 90}$ or $BN = 10\sin 30$ or $FN = 10\sin 60 = 5\sqrt{3}$ and $BN^2 = 10^2 - (5\sqrt{3})^2$ (may use any single letter instead of BN)
	A1 cso	$BN = 5$ (cm) Correct answer from correct working. (Answer given.) Must show a calculation in the form $BN = \ldots$ e.g. $BN = 10\cos 60$
(b)	M1	Use Pythag or trig to find a numerical expression for FN or FN^2 eg $FN^2 = 10^2 - 5^2$ $(= 75)$ or $FN = 10\sin 60$ $(= 5\sqrt{3} = 8.66...)$ or $FN = 5\tan 60$
	M1	Use Pythag with their FN to find a numerical expression for NE, eg $NE = \sqrt{12^2 + (5\sqrt{3})^2}$ $(= \sqrt{219})$
	A1	$NE = 14.8$ (cm) **given to 3SF**.
(c)	M1	Use tangent with their FN, eg $\tan RPN = \left(\dfrac{\frac{1}{2}\sqrt{75}}{12}\right)$ or any complete method.
	A1	19.8° **given to 1DP**.

Type MS · Angle between two planes using a right-angled triangle

Answer 3 of 11 · Question 8 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(d)	M1	Complete method to find AY eg $AY = \sqrt{(12^2 + 6.5^2) + \left(\frac{\sqrt{75}}{2}\right)^2}$ $(= \sqrt{205} = 14.31...)$ (allow their FN)
	A1	$AY = \sqrt{205}$ $(= 14.31\ldots)$ (may be implied by subsequent working)
	M1	Complete method to find FY eg $FC = \sqrt{75+3^2}$ $\left(= \sqrt{84}\right) \Rightarrow FY = \frac{1}{2}\sqrt{84}$ $(= \sqrt{21})$ (allow their FN) eg $FC = \sqrt{8^2 + 10^2 - 2\times 8\times 10\cos 60}$ $\left(= \sqrt{84}\right) \Rightarrow FY = \frac{1}{2}\sqrt{84}$ $(= \sqrt{21} = 4.58...)$
	M1	Method to find EY using their FY (correct or from a correct method) eg $EY = \sqrt{12^2 + 21}$ $(= \sqrt{165} = 12.84...)$
	M1	Substitute their values for AY and EY into the cosine rule used in triangle AYE. $\cos AYE = \left(\frac{205 + 165 - 100}{2\times\sqrt{205}\times\sqrt{165}}\right)$ oe
	A1	$\angle AYE = 42.8^\circ$ **given to 1DP**. Allow more than 1DP if 1DP was penalised in part (c).

Type MS · Angle between two planes using a right-angled triangle

Answer 3 of 11 · Question 8 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 4 of 11 · Question 9 part c · 2 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
9 (a)	$\cos\alpha = \frac{3}{\sqrt{13}}$	B1 (1)
(b)	$h = \sqrt{17^2 - \left(9^2 + 12^2\right)} = 8$	M1 M1 A1 (3)
(c)	Let M be the midpoint of BC $\tan\theta^\circ = \frac{h}{OM} = \frac{8}{12} = \frac{2}{3}$ *	M1 A1 (2)
(d)	Let N be the midpoint of EM	
	$EM = \sqrt{8^2 + 12^2} = 4\sqrt{13}$	M1
	$NO = \sqrt{12^2 + \left(2\sqrt{13}\right)^2 - 2(12)\left(2\sqrt{13}\right)\left(\frac{3}{\sqrt{13}}\right)} \Rightarrow NO = 2\sqrt{13}$	M1
	Hence triangle ONM is isosceles	
	$180 - 2\times 33.7 = 112.6^\circ$	M1 A1 (4)
		Total 10 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 4 of 11 · Question 9 part c · 2 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 5 of 11 · Question 9 part d · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

Number		
9 (a)	$\cos\alpha = \frac{3}{\sqrt{13}}$	B1 (1)
(b)	$h = \sqrt{17^2 - \left(9^2 + 12^2\right)} = 8$	M1 M1 A1 (3)
(c)	Let M be the midpoint of BC $\tan\theta^\circ = \frac{h}{OM} = \frac{8}{12} = \frac{2}{3}$ *	M1 A1 (2)
(d)	Let N be the midpoint of EM	
	$EM = \sqrt{8^2 + 12^2} = 4\sqrt{13}$	M1
	$NO = \sqrt{12^2 + \left(2\sqrt{13}\right)^2 - 2(12)\left(2\sqrt{13}\right)\left(\frac{3}{\sqrt{13}}\right)} \Rightarrow NO = 2\sqrt{13}$	M1
	Hence triangle ONM is isosceles	
	$180 - 2\times 33.7 = 112.6^\circ$	M1 A1 (4)
		Total 10 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 5 of 11 · Question 9 part d · 4 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 6 of 11 · Question 6 part whole · 6 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

6	Correctly identifies the angle VXO where X is the midpoint of CD and O is the foot of the perpendicular from V oe	B1
	If y is the length of AD. $VC = 3y$ $\quad CX = \frac{y}{2}$ $OX = \frac{3y}{2}$ $(VX) = \sqrt{(3y)^2 - \left(\frac{y}{2}\right)^2}$ $VX = \frac{\sqrt{35}y}{2}$ oe	M1 dM1 A1
	$\cos\theta = \frac{\frac{3y}{2}}{"\frac{\sqrt{35}y}{2}"}\left(= \frac{OX}{VC} = 0.507062.....\right)$ oe eg $\tan\theta = \frac{"\frac{\sqrt{26}}{2}y"}{\frac{3y}{2}}\left(= \frac{OV}{OX} = 1.699673....\right)$ $\sin\theta = \frac{"\frac{\sqrt{26}}{2}y"}{"\frac{\sqrt{35}y}{2}"}\left(= \frac{OV}{VC} = 0.861891.....\right)$ (Cosine rule) $\cos\theta = \frac{\left("\frac{\sqrt{35}y}{2}"\right)^2 + (3y)^2 - \left("\frac{\sqrt{35}y}{2}"\right)^2}{2\times\frac{\sqrt{35}}{2}y\times 3y}$ oe $\cos\theta = \frac{\left("\frac{\sqrt{35}y}{2}"\right)^2 + \left(\frac{3y}{2}\right)^2 - \left("\frac{\sqrt{26}y}{2}"\right)^2}{2\times\frac{\sqrt{35}}{2}y\times 3y}$ $\left[\cos\theta = \frac{3}{\sqrt{35}}\right.$ oe leading to $\theta =\left.\right]$ 59.5	M1 A1 [6]
		Total 6 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 6 of 11 · Question 6 part whole · 6 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
	B1	Angle identified in written work **or on diagram**. Allow labelling to be any letters. This mark should be permitted for any work where it is clear the candidate is identifying or using the correct angles. If ambiguous or unclear, cannot be awarded.
	M1	Denotes correct and appropriate letters for **any two** relevant lengths on the pyramid to make the next stage of progress. The sides are shown in the MS, but we will accept sides with different labels elsewhere in the triangle. Eg $DX = \frac{y}{2}$ is the same as CX This can be implied by written work or on the diagram. The two sides can be any **values** or expressions in the correct ratio which are necessary to form part of the working. Watch out for equivalences. Eg $AD = VC = y$ $DC = \frac{y}{3}$ $CX = \frac{y}{6}$ or $AD = VC = 9$ $DC = \frac{9}{3}$ $CX = \frac{9}{6}$ May be implied by correct working. For this mark, the sides used must have the correct algebraic expressions, (including multiples of those shown) or numbers in the correct proportion.
	dM1	Correct full method to find VX or VO. Correct full method means they can't use a side from an incorrect method in another method for VX or VO (so for example if they use an incorrect method to find VX and then use this to find VO they can't have the mark for finding VO) Dependent on the previous mark.
	A1	Correct expression **or values** for VO or VX ie all sides that they will be using.
	M1	Working in triangle VXO (or other valid triangle) with their values or expressions from previous working, using any appropriate trigonometry. The candidate can use sides which are labelled with the correct lengths for the trig ratio used, even if calculated incorrectly in previous work.
	A1	awrt 59.5 (59.52964053) Note: 60 is likely to have come from an incorrect angle identified And it is possible to get an answer close without a correct method. The method must always be checked through (in any question)

Type MS · Angle between two planes using a right-angled triangle

Answer 6 of 11 · Question 6 part whole · 6 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 7 of 11 · Question 11 part d · 3 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

11(a)	$AC = \sqrt{(16^2 + 16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)	
M1	Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1	Correct length for the diagonal or half diagonal
M1	Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso	Correct answer with no errors in the working
	OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
	$VP(= AP) = 8\sqrt{2}$ A1
	OR use any other complete valid method M1 Correct result A1
(b)	
M1	Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1	Correct numbers in their choice of method.
A1	Correct length obtained.
	$AV = 16$ w/o working scores M1A1A1

Type MS · Angle between two planes using a right-angled triangle

Answer 7 of 11 · Question 11 part d · 3 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

Type MS · Angle between two planes using a right-angled triangle

Answer 7 of 11 · Question 11 part d · 3 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

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Type MS · Angle between two planes using a right-angled triangle

Answer 8 of 11 · Question 4 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

4 (a)	$A = \frac{1}{2} \times 8 \times 8 \times \sin 60^\circ = \left(16\sqrt{3}\right)$ $48\sqrt{3} = \frac{1}{3} \times '16\sqrt{3}' \times h \Rightarrow h = 9$ *	M1 M1 A1cso [3]
(b)	$BX = \sqrt{9^2 + 8^2} = \sqrt{145}$ $\angle BXC = \frac{'145' + '145' - 8^2}{2 \times '\sqrt{145}' \times '\sqrt{145}'} = 38.8025...^\circ \approx 38.8^\circ$	M1 M1A1 [3]
(c)	Let midpoint of BC be M $AM = \sqrt{8^2 - 4^2} = \left(4\sqrt{3}\right)$ or $MX = \sqrt{145 - 4^2} = \left(\sqrt{129}\right)$ $\angle XMA = \tan^{-1}\left(\frac{9}{'4\sqrt{3}'}\right)$ or $\angle XMA = \sin^{-1}\left(\frac{9}{'\sqrt{129}'}\right)$ or $\angle XMA = \cos^{-1}\left(\frac{'4\sqrt{3}'}{'\sqrt{129}'}\right)$ $= 52.4109...^\circ \approx 52.4^\circ$	M1 M1 A1 [3]
	ALT 1	
	Let midpoint of BC be M $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$ $9^2 = \left(8^2 - 4^2\right) + \left(8^2 + 9^2 - 4^2\right) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$ $\theta = 52.4109...^\circ \approx 52.4^\circ$	{M1} {M1} {A1} [3]
		Total 9 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 8 of 11 · Question 4 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

ALT	
M1	Use of cosine rule using AX, AM and XM e.g. $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$
M1	Correct values substituted into the cosine rule in any form e.g. $9^2 = (8^2 - 4^2) + (8^2 + 9^2 - 4^2) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$
A1	awrt 52.4°

Type MS · Angle between two planes using a right-angled triangle

Answer 8 of 11 · Question 4 part c · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 9 of 11 · Question 6 part whole · 6 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

6	Correctly identifies the angle VXO where X is the midpoint of CD and O is the foot of the perpendicular from V There are other valid triangles that can be used.	B1
	$VC = a$ and $CX = \frac{a}{2}$ note could also be $2a$ and a or could use values where VC is twice the length of CX	M1
	$((VX)^2) = a^2 - \left(\frac{a}{2}\right)^2$	$((VO)^2) = \sqrt{a^2 - \left(\frac{\sqrt{2}}{2}a\right)^2}$	dM1
	$VX = \frac{\sqrt{3}a}{2}$ oe	$VO = \frac{\sqrt{2}}{2}a$ oe	A1
	$\cos\theta = \frac{\frac{a}{2}}{\frac{\sqrt{3}a}{2}}$ oe	$\tan\theta = \frac{\frac{\sqrt{2}a}{2}}{\frac{a}{2}}$ oe	M1
	$\left[\cos\theta = \frac{1}{\sqrt{3}}\text{oe}\right.$ leading to $\theta =]$	$[\tan\theta = \sqrt{2}$ leading to $\theta =]$	
	54.7	54.7	A1
			Total 6 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 9 of 11 · Question 6 part whole · 6 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angle between two planes using a right-angled triangle

Answer 10 of 11 · Question 3 part c · 4 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

3(a)		
	$AM = \sqrt{10^2 - 8^2}$, $= 6$	M1,A1 (2)
(b)		
	$\cos C = \frac{26^2+16^2-26^2}{2\times16\times26} = \frac{256}{832}$ $\left(=\frac{4}{13}\right.$ oe$\left.\right)$	M1A1
	$\angle BCD = 72°$	A1cao (3)
(c)	$AD = \sqrt{26^2 - 10^2} = 24$ or $DM = \sqrt{26^2 - 8^2} = 6\sqrt{17}$ oe (24.73...)	M1A1
	$\tan(\angle DMA) = \frac{24}{6}$ or $\cos(\angle DMA) = \frac{6}{6\sqrt{17}}$ or $\sin(\angle DMA) = \frac{24}{6\sqrt{17}}$	M1
	$\angle DMA = 76°$	A1cao (4) **[9]**

(a)	
M1	Use Pythagoras, with a minus sign, to obtain the length of AM.
A1	Correct length obtained.
NB	Answers w/o working get both marks (use of (3,4,5) triangle)
(b)	
M1	Use the cosine rule in ΔBCD to obtain a numerical expression for cos C. Correct formula in either form may be quoted and substitution attempted or correct formula can be implied by the correct substitution. Complete method needed, so if another angle found first do not award this mark until a value for angle BCD is obtained.
A1	Correct value for the cosine obtained (Decimal to be awrt 0.308 (0.30769...)) Award by implication if final answer is awrt 72°.
A1cao	For 72° (from correct working)
ALT	**Use the isosceles triangle**
M1A1	$\cos C = \frac{8}{26}$ oe (Any trig function allowed provided work completed to a value for angle BCD)
A1cao	For 72° (from correct working)
(c)	
M1	Attempt the length of AD or DM using Pythagoras with a minus sign.
A1	Correct value for their choice of line,
M1	Use an appropriate trig function. The length of AD or DM must have been attempted with a + or a - sign.
A1cao	Correct answer.
	Penalise once only in (b) and (c) for failing to round as instructed.

Type MS · Angle between two planes using a right-angled triangle

Answer 10 of 11 · Question 3 part c · 4 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$\longrightarrow$ $\longrightarrow$	

Type MS · Angle between two planes using a right-angled triangle

Answer 11 of 11 · Question 6 part c · 3 marks

2023 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

6(a)	$(AC=)\sqrt{x^2+x^2}\ (=\sqrt{2}x)$ **or** $(AC=)\dfrac{x}{\sin 45}$ **or** $(AC=)\dfrac{x}{\cos 45}$ $(AO=)\sqrt{\left(\dfrac{x}{2}\right)^2+\left(\dfrac{x}{2}\right)^2}\ \left(=\sqrt{\dfrac{x^2}{2}}=\dfrac{x}{\sqrt{2}}\right)$ **or** $(AO=)\dfrac{\frac{x}{2}}{\sin 45}$ **or** $(AO=)\dfrac{\frac{x}{2}}{\cos 45}$ $(VO=)\sqrt{x^2-\left(\dfrac{"\sqrt{2}x"}{2}\right)^2}$ **or** $\sqrt{x^2-\left(\dfrac{1}{2}\sqrt{x^2+x^2}\right)^2}$ **or** $\sqrt{x^2-\dfrac{x^2}{2}}\ \left(=\sqrt{\dfrac{1}{2}x^2}\right)$ **or** $(VO=)\dfrac{\sqrt{2}}{2}\tan 45$ $=\dfrac{\sqrt{2}}{2}x$ (cm)*	M1 M1 A1*cso [3]
ALT	Any indication the candidate realises triangle AVC is right angled and isosceles. $\sin 45=\dfrac{VO}{x}$ **or** $x\sin 45=VO$ $VO=\dfrac{\sqrt{2}}{2}x$ *	M1 M1 A1*cso [3]
(b)	$(\angle AVC=)\cos^{-1}\left(\dfrac{x^2+x^2-\left(\sqrt{2}x\right)^2}{2\times x\times x}\right)$ **or** $(\angle AVC=)2\times\sin^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\tan^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{\frac{\sqrt{2}}{2}x}\right)$ $=90^\circ$	M1 A1 [2]

Type MS · Angle between two planes using a right-angled triangle

Answer 11 of 11 · Question 6 part c · 3 marks

2023 NOV · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

(c)	Let M be the midpoint of AB and let N be the midpoint of DC	
	$(VM = VN =)\sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \left(\frac{\sqrt{3}}{2}x\right)$	M1
	$\cos MVN = \dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}$ **or**	
	$(\angle MVN =)\cos^{-1}\left(\dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}\right)$ **or**	M1
	$(\angle MVN =)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$ **or** $(\angle MVN =)2\times\sin^{-1}\left(\dfrac{\frac{1}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$	
	or $(\angle MVN =)2\times(\angle MVO =)$ **and** $\cos MVO = \dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}$	
	$70.528...^\circ \approx 70.5^\circ$	A1 [3]
	ALT	M1
	$OM = \frac{x}{2}$	
	$(\angle MVN =)2\tan^{-1}\left[\dfrac{\frac{x}{2}}{\frac{\sqrt{2}x}{2}}\right] = \left(2\tan^{-1}\left[\frac{1}{\sqrt{2}}\right] =\right)70.528...^\circ \approx 70.5^\circ$	M1A1 [3]
(d)	Volume $= \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2\left(=\frac{\sqrt{2}}{6}x^3\right)$ oe	M1
	$200 = \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2 \Rightarrow x =$	dM1
	$x = 9.4672... \approx 9.47(\text{cm})$	A1 [3]
		Total 11 marks

Type MS · Angle between two planes using a right-angled triangle

Answer 11 of 11 · Question 6 part c · 3 marks

2023 NOV · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
For part (a), we must see the algebraic proof, including x throughout.		
(a)	M1	For correct use of Pythagoras theorem or any correct appropriate trigonometry and a complete method to find the length AC or AO ie candidates must rearrange correctly – minimum steps as shown in mark scheme and no errors must be present.
	M1	For a full, correct and complete method to find VO, using their AC/AO.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions, minimum steps as shown.
ALT	M1	Any indication the candidate realises triangle AVC is right angled and isosceles. This mark may be implied by correct working for the next method mark or present on the diagram.
	M1	For the correct trig minimum steps as shown and no errors.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions.
(b)	M1	For any correct appropriate trigonometry to find angle AVC. $(\angle AVC =)$ If finding $\angle AVO$, must be doubled for this mark. AC must be $\sqrt{2}x$ oe, AO must be $\frac{\sqrt{2}}{2}x$ oe.
	A1	For 90°
	Allow M1A1 for candidates writing down the correct answer by inference.	
Consistent omission of x is permissible for all marks in part (b).		
(c)	M1	For correct Pythagoras theorem to find the lengths of VM or VN
	M1	For any correct trigonometry to find angle MVN, such as the statements shown in the mark scheme. The candidate can state, for example, cos(MVN) = fully correct expression (using their VN and VM) or, for example, $\cos^{-1}$(fully correct expression), where they do not have to write angle MVN = i.e – this can involve any fully correct method to reach an equation which would allow them to use inverse trig to find MVN or if they use inverse trig, a fully correct expression must be seen with the correct inverse trig function. Send any items where allocation of marks is unclear (potentially using other methods to the main and ALT) to Review. or fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
Consistent omission of x is permissible for all marks in part (c).		
ALT	M1	States or uses $OM = \frac{x}{2}$
	M1	For fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
(d)	M1	For a correct unsimplified expression for the volume of the pyramid as shown.
	dM1	For placing their expression = 200, which must involve a term in x^3, and a fully correct method/rearrangement to solve. Note this mark is dependent on the first method mark, so they must have got the correct unsimplified expression even if then simplified incorrectly.
	A1	For any final answer stated which rounds to 9.47 (ie more decimal places is condoned).

Type MS · Angle between two planes using a right-angled triangle

Answer 11 of 11 · Question 6 part c · 3 marks

2023 NOV · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks